LECTURE 4

Ex Determine if the following converge.

$\sum_{n=1}^{\infty} (-1)^n$ Divergence Test $\Rightarrow$ diverges

$\{ e^{-n} \}$ take limit $\Rightarrow$ converges to 0

Def'n $\sum_{n=1}^{\infty} \frac{1}{n}$ The <u>harmonic series</u> diverges

Def'n $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ The <u>alternating harmonic series</u> converges.

Def'n A series of the form $\sum_{n=0}^{\infty} ar^n$ is called a <u>geometric series</u>. The number r is called the <u>common ratio</u>.

Ex Find the value of a and r for the series $\frac{2}{3} + \frac{2}{9} + \frac{2}{27} + \frac{2}{81} + \dots$ and write in standard form.

We have $r = \frac{1}{3}$, $a = \frac{2}{3}$. Thus,

we have $\sum_{n=0}^{\infty} \frac{2}{3} \left(\frac{1}{3} \right)^n$

Ex Find a and r for the series $-\frac{2}{5} + \frac{4}{15} - \frac{8}{45} + \frac{16}{135} \dots$ and write in standard form

$a = -\frac{2}{5}$, $r = -\frac{2}{3}$ Thus,

we have $\sum_{n=0}^{\infty} -\frac{2}{5} \left(\frac{2}{3} \right)^n$

geometric series, common ratio

CONTINUED

Theorem

<u>Geometric Series Test</u>

For a geometric series $\sum_{n=l}^{\infty} ar^n$, if $|r| \geq 1$, then it diverges. If $|r| < 1$, then it converges to

$$\sum_{n=l}^{\infty} ar^n = \frac{ar^l}{1-r}$$

Ex

Determine if the following converge or diverge. If it converges, find the sum.

$$\sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n$$

This is a geometric series with $|r| = \frac{1}{3} < 1$, so it converges by the geometric series test.

$$\sum_{n=1}^{\infty} ar^n = \frac{1\left(\frac{1}{3}\right)^1}{1-\frac{1}{3}} = \frac{\frac{1}{3}}{\frac{2}{3}} = \frac{1}{2}$$

$$\sum_{n=6}^{\infty} \frac{(-2)^{3n}}{3^n}$$

We rewrite this as $\sum_{n=6}^{\infty} \left(\frac{-8}{3}\right)^n$

We have $|r| = \frac{8}{3} \geq 1$, so by the geometric series test it diverges.

$$\sum_{n=2}^{\infty} \frac{(-2)^n}{3^{2n+1}}$$

We rewrite this as $\sum_{n=2}^{\infty} \frac{1}{3}\left(\frac{-2}{9}\right)^n$.

It converges by the geometric series test since $|r| = \frac{2}{9} \leq 1$.

geometric series test

CONTINUED

Ex

$$\sum_{n=2}^{\infty} \frac{1}{3}\left(-\frac{2}{9}\right)^n = \frac{\frac{1}{3}\left(\frac{4}{81}\right)}{1-\left(\frac{-2}{9}\right)}$$

$$= \frac{4}{297}$$

Ex

Take a geometric series with $a = 1$ and $r = x$. We get a function $f(x) = \sum_{n=0}^{\infty} x^n$.

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots$$

By the geometric series test, it is when $|r| = |x| < 1$. Hence the domain is $-1 < x < 1$.

In the domain, we have $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$

Ex

If $\sum_{n=5}^{\infty} a_n$ converges, then does $\sum_{n=1}^{\infty} a_n$ converges.